

Scalability & Future Plan

Date	15 March 2026
Team ID	XXXXXX
Project Name	Rising water
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Single Point Sensor Deployment	System monitors only 1 location. Cannot detect floods in other areas	High
2	Reliance on Internet Connectivity	If network fails, real-time alerts and data sync to cloud will stop	High
3	Limited Battery Life of IoT Devices	Sensors need frequent recharging, not suitable for remote areas	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports 50 concurrent users on local server	Deploy on cloud with auto-scaling and load balancing
2	Data Storage	Stores 1 month of sensor data in MongoDB	Use cloud DB with archival policy. Add data compression
3	Performance	Data refresh every 5 seconds	Implement WebSockets + Redis caching for real-time updates
4	Security	Basic JWT authentication	Add role-based access, 2FA, and data encryption

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 1	Multi-Sensor Network Deployment	0-3 Months	Cover multiple locations in a city for area-wide flood detection
Phase 2	AI-Based Predictive Analytics	3-6 Months	Improve flood prediction accuracy to 95% using historical + weather data
Phase 3	Mobile App + Government Dashboard Integration	6-9 Months	Send direct alerts to citizens and disaster management teams